ProQuant ANALYTICS

INDEPENDENT STRATEGY VALIDATION · INVESTOR SUMMARY

Dual moving-average trend (long-only)

Demo Client · fast=20,slow=100 · 2013-01-01 to 2025-12-31 · 11 November 2026

Demonstration. Synthetic data generated for illustration; no real securities or client data.

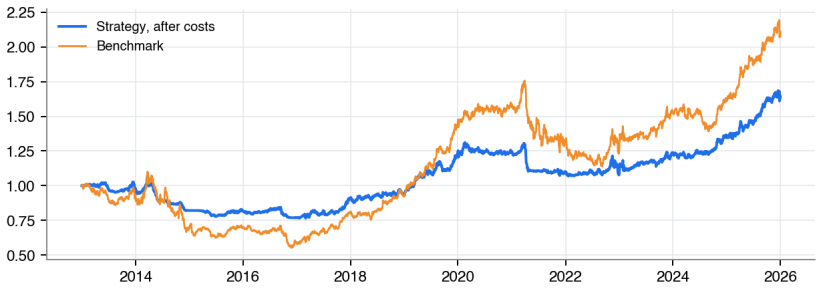
Validated with reservations - 2 point(s) to address

Independent assessment by ProQuant Analytics against pre-agreed validation thresholds (11 PASS, 2 WARN). Not a forecast or investment recommendation.

NET CAGR	NET SHARPE	VOLATILITY	MAX DRAWDOWN
3.8%	**0.46**	**8.9%**	**-28.9%**
gross 4.0%	benchmark 0.43	annualised	net of costs

DEFLATED SHARPE	PROB. OF OVERFITTING	WALK-FORWARD SHARPE	BREAK-EVEN COST
67.6%	**2.8%**	**1.24**	**97 bps**
45 trials; 95% required	CSCV; lower is better	out of sample	assumed 5.0 bps

Growth of 1 (net of costs) vs benchmark



Risk and return

CAGR	3.8%	5.7%
Sharpe	0.46	0.43
Sortino	0.63	0.60
Max drawdown	-28.9%	-49.9%
Calmar	0.13	0.11
VaR 95% (1 day)	0.8%	1.4%
ES 95% (1 day)	1.4%	2.4%
Beta to benchmark	0.45	n/a
Hit rate (days)	52.1%	52.1%

Columns: strategy (net) · benchmark

Calendar-year returns

	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Strategy (net)	-3.0%	-15.3%	-0.2%	-6.0%	17.4%	3.6%	32.5%	0.5%	-12.0%	-1.8%	14.2%	11.0%	20.6%
Benchmark	-8.0%	-24.5%	0.9%	-16.3%	36.9%	14.8%	59.3%	8.3%	-16.8%	-7.0%	23.7%	6.7%	29.2%

Validation checks

Data quality after cleaning	0	PASS
Look-ahead bias test	passed	PASS
Execution-delay robustness	0.98x	PASS
Survivorship bias	0.8%	PASS
Net Sharpe ratio	0.46	WARN
Sharpe with 2x costs	0.44	PASS
Deflated Sharpe Ratio	67.6%	WARN
Probability of overfitting	2.8%	PASS
Walk-forward efficiency	1.53x	PASS
Hold-out vs in-sample	6.62x	PASS
Parameter robustness	100.0%	PASS
Parameter plateau	0.84x	PASS
Monte Carlo P(Sharpe<0)	4.5%	PASS

Key points

- Net of costs: 3.8% a year, Sharpe 0.46, max drawdown -28.9%.
- Out of sample the edge held: walk-forward Sharpe 1.24, hold-out Sharpe 1.06.
- Costs are not the risk: break-even 97 bps per trade vs 5.0 bps assumed.
- Statistical confidence after 45 trials: DSR 67.6% (95% required).
- Monte Carlo 90% range for Sharpe: 0.02 to 0.93.

Prepared for Demo Client. Based on client-supplied data and the methodology in the full validation report, which should be read for assumptions and limitations. Simulated and past performance is not a reliable indicator of future results. Not investment advice.